

LEARNING RESOURCE GUIDE
ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional
Participant Workbook

Incident, Problem, and Continual Service Improvement Management

1. Learning Objectives

By the end of this lecture, you will be able to:

- Clearly distinguish incident management from problem management by purpose, time horizon, and output.
- Describe the incident management lifecycle including detection, classification, escalation, resolution, and closure.
- Explain what root cause analysis produces and how its outputs feed the change management process.
- Define availability management and capacity management and relate them to SLA commitments for cloud services.
- Explain Continual Service Improvement (CSI) as a structured feedback loop and describe its practical implementation requirements.

2. Introduction

At 2:14 AM, the monitoring system at a large SaaS company fires a Severity 1 alert: the primary database cluster for its European customer base has become unreachable. The on-call engineer wakes, connects, diagnoses a connection pool exhaustion issue, and restores the service by restarting the database proxy. The incident is resolved in 47 minutes. An entry is logged. The engineer goes back to sleep. Three weeks later, the same alert fires at 1:58 AM. Same diagnosis. Same fix. Same log entry. Four weeks after that: 2:31 AM, same alert.

The incident management process worked correctly each time. The service was restored within the SLA window. But the same incident recurring three times in seven weeks is not a success. It is the unmistakable signature of an operations function that resolves symptoms without addressing causes — a team running incident management in the absence of problem management.

This lecture covers the operational disciplines that distinguish a reactive team from a proactive one: incident management for fast service restoration, problem management for permanent root cause elimination, availability and capacity management for honoring service commitments before failures occur, and Continual Service Improvement for turning operational experience into systematic program progress. These are not competing processes. They are a connected lifecycle, and the strength of any cloud security operations function is measured by how well they reinforce each other.

3. Core Concepts Explained

3.1 Incident Management

Definition: Incident management is the process of restoring normal service operation as quickly as possible following an unplanned interruption or degradation of service quality. Its defining characteristic is that speed is the primary objective. Incident management does not pause to understand why a failure occurred. That is problem management's job. Incident management finds the fastest path back to a working system and takes it.

Business Relevance: SLA commitments — particularly Recovery Time Objectives (RTOs) and mean time to restore (MTTR) targets — are measured against the incident management process. A slow or disorganized incident response directly translates to SLA breaches, which trigger financial penalties and customer trust erosion. For cloud services processing financial transactions, healthcare data, or real-time communications, every additional minute of degraded service has a measurable dollar cost.

Incident Lifecycle: Detection is the first phase — the incident must be identified before it can be addressed. Effective detection requires monitoring coverage across all service layers: infrastructure metrics, application error rates, security event streams, and synthetic transaction monitoring. Classification assigns a severity level and impact category. Severity levels must be defined in advance, not during the incident: a team debating what 'Severity 2' means while a service is degraded will respond slowly and inconsistently. Escalation routes the incident to the team with the authority and skills to resolve it. Resolution implements the fastest available fix, which may be a restart, a configuration rollback, a traffic failover, or a network route change. Closure documents the incident record — including timeline, actions taken, resolution steps, and systems affected.

Cloud-Specific Considerations: Cloud environments introduce shared responsibility complexities into incident management. When a production database service fails, the cloud security team must quickly determine whether the failure is in the cloud provider's managed service layer (requiring engagement with the CSP's support) or in the customer's application configuration or network routing (requiring internal resolution). Pre-defined escalation paths that cover both internal teams and CSP support channels are essential.

Real-World Example: When AWS experienced the US-EAST-1 region disruption in December 2021, the incident impacted thousands of dependent services globally. AWS's incident management process was publicly visible through their Service Health Dashboard, which posted status updates approximately every 30 minutes. The structured communication — acknowledging the incident, providing scope, estimating recovery timeline, and confirming resolution — is the customer-facing artifact of a mature incident management process. Organizations without their own documented incident management runbooks for CSP outage scenarios were slower to activate their continuity responses.

Risks: Severity classification that is too coarse — defining everything as Severity 1 creates alert fatigue; defining too few things as Severity 1 causes under-response. Incident

management without post-closure analysis that feeds into problem management — restoring service without investigating cause condemns the team to repeat incidents.

3.2 Problem Management

Definition: Problem management is the discipline that identifies the underlying causes of incidents and drives permanent fixes. Where incident management asks 'How do we restore service now?', problem management asks 'Why did this happen, and how do we prevent it from happening again?' A problem is the unknown underlying cause of one or more incidents. A known error is a problem that has been investigated to the point where the root cause is understood, even if a permanent fix has not yet been implemented.

Business Relevance: Without problem management, an operations function accumulates technical debt in the form of recurring incidents. Each recurrence consumes on-call resources, erodes customer confidence, and creates regulatory exposure if the affected service is subject to availability SLA commitments under a customer contract. Problem management converts reactive fire-fighting into a managed program of root cause elimination.

Practitioner Angle: Root cause analysis (RCA) is the primary tool of problem management. After a significant incident — or after a cluster of lower-severity incidents with a shared pattern — the problem management process initiates an RCA to identify the structural condition that made the failure possible. The RCA output is not a blame assignment; it is a technical finding that produces a change request to modify the system so the failure mechanism is eliminated or its probability and impact are reduced. This change request then enters the standard change management process, creating a direct handoff between problem management and change management.

The most mature cloud operations teams publish post-mortem reports as a cultural practice. A well-written post-mortem describes what happened, the timeline of the incident and the response, contributing factors at multiple system layers, the root cause finding, and the specific action items being taken to prevent recurrence. Post-mortems are blameless by design — the goal is to understand and fix the system, not to find a person responsible for pressing the wrong button.

Real-World Example: Google has published its approach to post-mortem culture in its Site Reliability Engineering book and public blog posts. After significant service disruptions, Google's SRE teams conduct structured blameless post-mortems and publish the findings publicly in their cloud status page post-incident summaries. Each published summary includes not just the what and why but the specific infrastructure or process changes being made to prevent recurrence. This is problem management applied at hyperscale — systematic root cause elimination treated as a first-class engineering discipline rather than an afterthought.

Risks: Post-mortems that produce action items without owners or deadlines — the findings sit in a document and the system never changes. Treating every incident as a unique

random event rather than looking for patterns across incident clusters. Conducting RCA only for the highest-severity incidents while ignoring the medium-severity recurring ones that accumulate the most aggregate impact.

3.3 Availability Management and Capacity Management

Definition: Availability management is the practice of ensuring that IT services meet agreed availability requirements. In cloud contexts, this means designing for the redundancy, fault tolerance, and failover capability needed to honor uptime commitments. Capacity management ensures that the resource capacity — compute, storage, network throughput, database connections — needed to sustain service levels is available before demand arrives, not after it spikes.

Business Relevance: Service Level Agreements quantify availability and performance commitments. A 99.9 percent uptime SLA allows approximately 8.7 hours of downtime per year. A 99.99 percent SLA allows about 52 minutes. These are not just marketing numbers — they translate directly into financial penalties, contract termination rights, and regulatory findings when breached. Availability management and capacity management are the engineering disciplines that make SLA commitments achievable rather than aspirational.

Practitioner Angle: Availability management works at design time (building redundant architectures with no single points of failure), at operational time (monitoring availability metrics against SLA thresholds in real-time), and at failure time (ensuring failover mechanisms actually function when needed through regular DR testing). Capacity management works on predictive modeling: examining historical usage trends and scheduled demand spikes — Black Friday traffic patterns, end-of-quarter reporting batch jobs, new product launches — and ensuring resources are provisioned in advance. Cloud auto-scaling can handle unexpected demand within bounds, but capacity planning defines those bounds.

Real-World Example: In August 2016, Delta Air Lines suffered a massive IT outage that grounded flights for two days and cost an estimated \$150 million. The root cause was traced to a power control module failure at their primary data center that was not adequately covered by redundancy. The failure triggered a cascading collapse of critical systems including crew scheduling, passenger check-in, and flight operations systems. Availability management failure at the infrastructure design layer — insufficient redundancy for a critical control component — led to an outage that far exceeded any reasonable RTO. The incident drove Delta to significantly overhaul its availability architecture with cloud migration as a core component.

3.4 Continual Service Improvement

Definition: Continual Service Improvement (CSI) is the structured, ongoing discipline of using operational data — incident trends, SLA performance, problem management findings, capacity metrics, and customer feedback — to systematically improve service quality over

time. CSI is not a one-time audit or an annual review. It is a feedback loop that runs on a regular cadence.

Business Relevance: Organizations that implement CSI effectively become progressively more reliable over time. MTTR decreases as incident response runbooks improve. Recurring incident rates decline as problem management eliminates root causes. SLA compliance rates increase as availability management designs improve. CSI converts the operational experience that accumulates in every cloud environment into compounding improvements rather than allowing it to evaporate after each incident closure.

Practitioner Angle: CSI requires four operational enablers. First, a named owner — CSI without accountable ownership produces meeting minutes, not change. Second, a regular cadence — monthly or quarterly service review meetings where metrics from the past cycle are presented, trended over time, and compared to targets. Third, a visible record — an improvement register or backlog where action items are tracked with owners, deadlines, and status. Fourth, closed-loop verification — confirming that completed actions actually produced the improvement their authors predicted, rather than assuming that action completion equals improvement achievement.

Real-World Example: Microsoft's Azure team publishes detailed post-incident reviews (PIRs) following significant Azure service events. Beyond describing the incident, these PIRs explicitly list the 'next steps' the Azure team is taking — architectural changes, deployment safeguards, monitoring improvements, and process changes. Tracking these PIR action items across Azure's published history shows a continual improvement discipline in action: the same failure mode rarely produces the same Azure disruption twice, because the problem management and CSI loop actually closes. This is what distinguishes a mature cloud operations function from a reactive one.

Risks: CSI programs that generate improvement action items but track no metrics to confirm the improvements are working. Review meetings that become reporting sessions without decision-making — metrics are presented but no one is empowered to change the operations that produced them. Using CSI to document improvements that have not yet been implemented, creating a compliance artifact rather than an operational reality.

4. Incident Management vs. Problem Management

Dimension	Incident Management	Problem Management
Primary Goal	Restore service as fast as possible	Identify and eliminate root causes permanently
Time Horizon	Minutes to hours (reactive, during the event)	Days to weeks (retrospective, after

		closure)
Key Output	Restored service; incident record	Root cause finding; change request for permanent fix
Key Metric	Mean Time to Restore (MTTR)	Recurring incident rate; known error count
Success Looks Like	SLA met; service back online	The same incident stops recurring
ITIL Concept	Incident management practice	Problem management practice; known error record

4.1 SLA Availability Calculations

Availability Target	Max Annual Downtime	Max Monthly Downtime	Architecture Implication
99.0% (two nines)	~87.6 hours	~7.3 hours	Single-region with failover
99.9% (three nines)	~8.7 hours	~43.8 minutes	Multi-AZ; auto-failover
99.95%	~4.4 hours	~21.9 minutes	Active-active multi-AZ; fast DNS failover
99.99% (four nines)	~52.6 minutes	~4.4 minutes	Multi-region active-active; global load balancing
99.999% (five nines)	~5.3 minutes	~26 seconds	Full geo-redundancy; zero-RPO/RTO design

5. Real-World Incident

Incident: On July 8, 2015, United Airlines grounded all domestic flights for approximately two hours due to a computer networking issue — later identified as a router configuration problem. On the same day, the New York Stock Exchange halted trading for nearly four hours due to a software deployment issue, and the Wall Street Journal website went offline briefly. While these three events were unrelated and were confirmed not to be a coordinated cyber-attack, they illustrated a common operations failure: all three were recoverable system failures that were not resolved quickly because incident escalation paths, severity definitions, and response runbooks had not been sufficiently tested or kept current. United's incident management process eventually restored operations but took longer than it should have given the nature of the problem. The NYSE outage was later

attributed to a software deployment that introduced a configuration mismatch — a failure that effective change management and deployment validation would have prevented, and that a faster incident management process would have resolved sooner. Together, the events provided a high-profile illustration of why incident classification, escalation, and continual improvement of response procedures matter for operational resilience.

6. Memory Anchor

Think of incident management, problem management, and continual service improvement as the three-stage response to a fire in a building. Incident management is the fire department: their job is to put out the fire as fast as possible. They do not stop during the fire to investigate why the wiring was faulty. They extinguish, evacuate, and restore safety. Problem management is the fire investigation team: they arrive after the fire is out, examine the scene, and determine the root cause — faulty wiring in a specific wall cavity connected to an overloaded circuit. Their report generates a building code correction request (the change management handoff). Continual service improvement is the building inspector who reviews the fire investigation reports across all buildings every six months, identifies patterns (faulty wiring is the most common root cause in buildings older than 30 years), and updates the inspection program to catch this risk proactively. The fire department can do its job perfectly every time and the building still burns repeatedly — unless the investigation team and the inspector are also doing their jobs.

7. Exam Focus

The CCSP exam's most common question type for this topic is the incident/problem distinction. Incident management restores service — it does not investigate root cause. Problem management identifies root cause and drives permanent fixes. When a scenario asks about investigating why incidents keep recurring, the answer is problem management. When a scenario asks about restoring service within the SLA window, the answer is incident management.

A second key distinction is problem vs. known error. A problem is an unresolved root cause — the 'why' is not yet understood. A known error is a problem where the root cause has been identified, even if the permanent fix has not yet been implemented. The distinction matters because known error records allow incident management teams to apply workarounds quickly for recurring incidents while the permanent fix is in progress.

Availability management scenarios involve SLA calculations, uptime targets, and architecture design for redundancy. If a question gives you an uptime percentage and asks what it implies about maximum downtime, work the math: $(1 - \text{availability}\%) \times \text{hours in year}$. For SLA breach consequence scenarios, the answer is almost always that the breach triggers financial penalties or contract remedies.

CSI scenarios test whether the candidate understands that CSI is a continuous feedback loop, not a one-time event. When a scenario asks how an organization ensures that the same

service improvement actually achieved the intended result, the answer involves CSI's closed-loop verification mechanism — tracking metrics after the improvement is implemented to confirm the outcome, not just completing the action item.

Common distractors: confusing MTTR (a metric for incident management performance) with MTBF (mean time between failures — a metric for availability planning). Selecting 'change management' when a question asks how recurring incidents are permanently eliminated — change management implements the fix, but problem management identifies what to fix. Treating post-mortem reports as a compliance document rather than a problem management output that must drive change.

8. Key Terms Glossary

- **Incident:** An unplanned interruption or reduction in quality of an IT service requiring restoration.
- **Incident Management:** The process focused on restoring normal service operation as quickly as possible following an incident.
- **Severity Classification:** Pre-defined categories (e.g., Severity 1-4) determining response urgency, escalation path, and resource priority for incidents.
- **Problem:** The unknown underlying cause of one or more incidents — under investigation by problem management.
- **Known Error:** A problem where the root cause has been identified; a workaround may be available while a permanent fix is developed.
- **Root Cause Analysis (RCA):** The investigation process that identifies the structural condition that made a failure possible, producing findings that drive permanent remediation.
- **Post-Mortem (Blameless):** A structured, blame-free review of a significant incident documenting timeline, contributing factors, root cause, and specific remediation actions.
- **Mean Time to Restore (MTTR):** The average time to restore service after an incident — the primary metric for incident management effectiveness.
- **Mean Time Between Failures (MTBF):** The average time between service failures — a metric for availability planning and system reliability.
- **Service Level Agreement (SLA):** A formal commitment specifying minimum acceptable service performance including uptime, response time, and MTTR targets.
- **Availability Management:** The practice of designing, monitoring, and maintaining IT services to meet agreed availability commitments.
- **Capacity Management:** The practice of ensuring sufficient resource capacity is provisioned to meet service demand before demand spikes occur.
- **Continual Service Improvement (CSI):** The structured feedback loop that uses operational data to systematically improve service quality over time.

9. Quick Revision Checklist

- ☐ I can distinguish incident management from problem management by purpose, time horizon, output, and key metric.

- ☐ I can describe the five phases of incident management: detect, classify, escalate, resolve, close.
- ☐ I understand why severity levels must be defined before an incident — not during one.
- ☐ I can define 'problem' and 'known error' and explain how they relate to the RCA process.
- ☐ I can explain how RCA outputs feed the change management process to produce permanent fixes.
- ☐ I can calculate maximum annual downtime from an uptime percentage (e.g., 99.9% = 8.76 hours/year).
- ☐ I can explain the difference between availability management (design/monitor for uptime) and capacity management (provision resources before demand).
- ☐ I can describe CSI's four requirements: named owner, regular cadence, visible improvement register, closed-loop verification.
- ☐ I know the MTTR/MTBF distinction and which process each metric primarily belongs to.
- ☐ I can name the AWS December 2021 US-EAST-1 outage and the Delta 2016 outage as real-world availability management examples.

10. Practice Questions

Question 1 (Scenario)

A cloud operations team at a fintech company has been responding to the same database connectivity incident for the fourth time in eight weeks. Each time, the on-call engineer restores the service in approximately 40 minutes by restarting the database connection pool. The SLA requires restoration within 60 minutes, so the team is meeting its SLA each time. The CISO reviews the incident log and asks why the engineering team has not addressed the underlying issue. Which process is the team failing to execute?

A. Incident management, because they are not meeting the SLA window.

Explanation: Incorrect. The team is meeting the SLA — restoration occurs in 40 minutes against a 60-minute target. The failure is not in incident management performance; it is the absence of a parallel process to address the underlying cause.

B. Problem management, because recurring incidents indicate an unresolved root cause that no one is investigating.

Explanation: Correct. The same incident recurring four times in eight weeks is a textbook indicator of a root cause that has never been identified or fixed. Problem management is the discipline responsible for investigating why incidents recur and driving permanent remediation through the change management process.

C. Change management, because the team should have raised an RFC to fix the database configuration.

Explanation: Incorrect. Change management implements fixes, but the fix cannot be designed until problem management identifies the root cause. The failure is upstream: the root cause investigation has not been initiated, so there is nothing for change management to implement yet.

D. Capacity management, because the database connection pool is running out of connections due to insufficient provisioning.

Explanation: Incorrect. While connection pool exhaustion could be a capacity issue, the question does not provide enough information to conclude this is the root cause. The identified failure is the absence of root cause investigation — whatever the root cause turns out to be, problem management is the missing process.

Question 2 (Scenario)

During a major cloud service incident, the on-call team identifies a workaround — routing traffic through a secondary load balancer — that restores service within 35 minutes. The underlying cause of the primary load balancer failure is not yet understood. The incident is closed with the workaround applied and a record noting that root cause is unknown. Three days later, a problem management review is initiated. What is the correct ITIL terminology for the state of this issue when the problem management review begins?

A. Known error, because the incident has been resolved and documented.

Explanation: Incorrect. A known error is a problem where the root cause has been identified. At the start of the problem management review, the root cause is explicitly unknown. The incident resolution with an unknown cause defines this as a problem, not yet a known error.

B. Problem, because the root cause of the load balancer failure has not yet been identified.

Explanation: Correct. In ITIL terminology, a problem is an issue where the underlying cause of one or more incidents is unknown. Because the root cause of the load balancer failure has not been identified, it is a problem. It becomes a known error once the RCA identifies the root cause.

C. Standard change, because the team applied a workaround during the incident.

Explanation: Incorrect. A standard change is a pre-approved change template for routine activities. Applying a workaround during an incident is an incident management action, not a change type. The workaround may later generate a change request through problem management, but the workaround itself is not classified as a change type.

D. Service request, because the operations team requested routing through the secondary load balancer.

Explanation: Incorrect. A service request in ITIL is a user or customer request for standard service items such as account provisioning. Traffic rerouting during an incident is an incident response action, not a service request.

Question 3 (Scenario)

A SaaS provider's enterprise contract commits to 99.99 percent monthly availability for its primary application platform. Over the past month, the platform experienced two unplanned outages: one lasting 12 minutes and one lasting 45 minutes, for a combined downtime of 57 minutes. The provider's operations team must assess whether they have breached the SLA. What is the maximum allowable monthly downtime under the 99.99 percent commitment, and has the SLA been breached?

A. Maximum monthly downtime is approximately 4.4 minutes; the SLA has been breached.

Explanation: Correct. A 99.99 percent monthly availability commitment allows $(1 - 0.9999) \times 43,800$ minutes per month = approximately 4.38 minutes of downtime per month. With 57 minutes of actual downtime, the provider has breached the SLA significantly. The breach would typically trigger financial penalties or service credits under the contract.

B. Maximum monthly downtime is approximately 52 minutes; the SLA has not been breached.

Explanation: Incorrect. Approximately 52 minutes per month corresponds to 99.99 percent annual availability ($8,760 \text{ hours} \times 0.0001 \times 60 = 52.56$ minutes per year), not monthly. The monthly calculation at 99.99 percent yields approximately 4.4 minutes.

C. Maximum monthly downtime is approximately 43.8 minutes; the SLA has been breached.

Explanation: Incorrect. Approximately 43.8 minutes per month corresponds to 99.9 percent (three nines) monthly availability, not 99.99 percent. At 99.99 percent monthly, the allowance is approximately 4.4 minutes.

D. Maximum monthly downtime is approximately 8.7 hours; the SLA has not been breached.

Explanation: Incorrect. 8.7 hours corresponds to 99.9 percent annual availability. The 99.99 percent monthly calculation yields approximately 4.4 minutes — far less than 8.7 hours.

Question 4

What is the primary objective that distinguishes incident management from all other ITSM processes?

A. Identifying the root cause of service failures to prevent recurrence.

Explanation: Incorrect. Root cause identification is the objective of problem management. Incident management deliberately does not pause to investigate cause — its defining objective is speed of restoration.

B. Restoring normal service operation as quickly as possible.

Explanation: Correct. Incident management's defining and primary objective is service restoration speed. All other considerations — root cause investigation, trend analysis, process improvement — are the domain of other processes. This single-minded focus on speed is what makes incident management effective.

C. Documenting the change that caused the incident for audit purposes.

Explanation: Incorrect. Documentation of the causative change is part of problem management's RCA and feeds change management. Incident management documents the incident record but its primary purpose is not documentation — it is restoration.

D. Ensuring the incident does not recur by implementing a permanent fix.

Explanation: Incorrect. Permanent fix implementation is the output of problem management (which identifies the root cause) feeding into change management (which implements the fix). Incident management restores service; it does not eliminate root causes.

Question 5

A cloud operations center defines its incident severity levels as: Severity 1 — complete service outage; Severity 2 — major functional impairment affecting more than 30% of users; Severity 3 — minor functional impairment. A new engineer joins the team and during their first on-call shift, encounters an alert that doesn't clearly match any definition. They spend 12 minutes researching the severity level before taking action. What is the fundamental process design failure this scenario illustrates?

A. The team should use problem management to define severity levels.

Explanation: Incorrect. Severity level definitions are a change management and service management design decision, not a problem management output. Problem management addresses root causes of incidents, not classification framework design.

B. Severity definitions should be unambiguous, complete, and trained-against before an incident occurs, not discovered during one.

Explanation: Correct. The 12-minute delay in classification represents response time lost to a process design failure: severity levels that are insufficiently clear or have gaps. Classification frameworks must cover all common alert types, be trained against regularly,

and be posted in every runbook so that any engineer can classify accurately within seconds of encountering an alert.

C. The team should have implemented automation to classify all alerts without human judgment.

Explanation: Incorrect. While automation can assist with alert triage, human judgment is often needed for edge cases. The failure is not absence of automation — it is inadequate preparation of the human responder through training and clear classification criteria.

D. The engineer should have escalated the unclassified alert to their manager immediately.

Explanation: Incorrect. Escalating unclassified alerts without classification guidance does not solve the underlying problem — the next engineer will face the same ambiguity. The solution is fixing the classification framework, not routing the decision upward every time.

Question 6

Root cause analysis (RCA) is the primary tool of problem management. What is the correct sequence of outputs from an RCA finding to permanent remediation?

A. Incident record → Known error → Change request → Permanent fix

Explanation: Incorrect. The incident record is created during incident management, not as part of the RCA sequence. The RCA begins from a problem record (opened after the incident), not from the incident record directly. The sequence is: problem record → RCA → known error → change request → permanent fix.

B. Problem record → RCA finding → Known error record → Change request to change management → Permanent fix implemented

Explanation: Correct. After incident closure, a problem record is opened. Problem management conducts an RCA. Once the root cause is understood, the problem becomes a known error. The known error record generates a change request that enters the change management process. The approved change implements the permanent fix.

C. Escalation → RCA → Emergency change → Problem closure

Explanation: Incorrect. Escalation is an incident management step. Emergency changes are for urgent situations, not the typical output of problem management RCA. The standard output is a normal or standard change request following the full CAB process.

D. SLA breach → Availability review → Capacity adjustment → Problem closure

Explanation: Incorrect. An SLA breach triggers availability management review, which is a separate process from problem management's RCA. Capacity adjustment addresses resource provisioning. Problem management RCA addresses root cause elimination, which may or may not be a capacity issue.

Question 7

A cloud services company is preparing for a significant marketing campaign that is expected to triple its normal peak traffic for a 72-hour period. The operations team must ensure that the platform can sustain the load without performance degradation. Which discipline is primarily responsible for this preparation?

A. Incident management

Explanation: Incorrect. Incident management is reactive — it responds to failures after they occur. The question describes a proactive pre-campaign activity to prevent failures, which is the domain of capacity management.

B. Problem management

Explanation: Incorrect. Problem management addresses root causes of existing incidents. The marketing campaign scenario is a forward-looking provisioning exercise, not a response to a recurring failure pattern.

C. Capacity management

Explanation: Correct. Capacity management ensures that the resources needed to meet service demand are available before demand arrives. Pre-provisioning for a known traffic spike — scaling compute, increasing database connection pools, expanding CDN capacity — is the practical expression of capacity management for a campaign event.

D. Continual Service Improvement

Explanation: Incorrect. CSI uses historical operational data to drive ongoing service improvements. While it might inform campaign preparation through trend analysis, the specific pre-campaign provisioning activity is capacity management.

Question 8

Which of the following best describes a blameless post-mortem in the context of problem management?

A. A review meeting where the incident timeline is reconstructed and the engineer who made the causative change is formally counseled.

Explanation: Incorrect. A blameless post-mortem explicitly avoids blame assignment to individuals. Its purpose is to understand systemic conditions that allowed the failure, not to identify and penalize the individual who triggered it.

B. A structured review that identifies contributing factors, root cause, and specific system or process changes that will prevent recurrence, without attributing fault to individuals.

Explanation: Correct. The blameless post-mortem's defining characteristic is its focus on the system rather than the people. Humans make mistakes; blameless post-mortems ask why the system allowed a human mistake to produce a catastrophic outcome, and what changes to the system will prevent the same outcome next time.

C. A review conducted only for incidents classified as Severity 1.

Explanation: Incorrect. While Severity 1 incidents always warrant post-mortems, blameless post-mortems can and should be conducted for significant Severity 2 and recurring lower-severity incidents. Restricting post-mortems to Severity 1 events misses the patterns that medium-severity recurring incidents reveal.

D. A document produced by the legal team after an SLA breach to prepare the company's defense against customer claims.

Explanation: Incorrect. Post-mortems are engineering and operations documents produced to improve service reliability. They are not legal instruments, and their production is not primarily driven by SLA breach scenarios.

Question 9

What is the primary risk when an organization implements Continual Service Improvement (CSI) by collecting metrics and producing action item lists, but takes no further governance steps?

A. The CSI process will conflict with incident management by diverting on-call engineers to improvement tasks.

Explanation: Incorrect. CSI and incident management operate on different timescales and should not compete for the same resources. CSI runs on a regular review cadence; incident management is reactive. Resource conflict is a management scheduling concern, not a primary CSI governance risk.

B. Improvement action items without named owners, deadlines, and closed-loop verification will produce documentation artifacts rather than actual service improvement.

Explanation: Correct. The most common CSI failure mode is generating improvement lists that are never executed or never verified. Without accountability (named owner, deadline) and validation (confirming the improvement achieved its intended metric outcome), CSI becomes a compliance theater exercise that does not produce better service.

C. The CSI process will interfere with problem management by opening RCAs for issues that have already been resolved.

Explanation: Incorrect. CSI and problem management are complementary: problem management RCA findings should feed CSI improvement actions. They do not conflict; they reinforce each other in a well-run operations function.

D. Collecting too many metrics will overwhelm the operations team and reduce incident response speed.

Explanation: Incorrect. Metrics collection, if automated, does not inherently burden on-call engineers. The primary CSI risk is not metric overload — it is the failure to act on the metrics that are collected.

Question 10

A CSP's monthly SLA report shows that its primary compute platform has had 99.97 percent availability over the past quarter. Its contractual commitment is 99.99 percent per month. Which statement about the SLA status over this quarter is accurate?

A. The CSP has met its SLA because 99.97 percent is above 99 percent.

Explanation: Incorrect. SLA thresholds are specific contractual values, not general tiers. The commitment is 99.99 percent; 99.97 percent falls below this threshold regardless of its absolute value. Meeting '99 percent' is irrelevant when the commitment is '99.99 percent.'

B. The CSP has breached its SLA, as 99.97 percent is below the 99.99 percent contractual commitment, and financial penalties or service credits likely apply.

Explanation: Correct. 99.97 percent is 0.02 percentage points below the 99.99 percent commitment. Monthly, this translates to approximately 13.1 minutes of excess downtime against the 4.4-minute allowance. The SLA has been breached, and most enterprise SLA contracts specify service credits or penalty clauses triggered by such breaches.

C. The SLA status cannot be determined from quarterly data — only monthly measurements determine SLA compliance.

Explanation: Incorrect. While SLA compliance is typically measured monthly, if monthly performance data is available within the quarterly report, each month's SLA compliance can be assessed. The quarterly average of 99.97 percent being below 99.99 percent indicates that at least some months in the quarter breached the SLA.

D. The CSP is within SLA tolerance because industry standard allows a 0.05 percent deviation from stated commitments.

Explanation: Incorrect. SLA terms are specific contract provisions, not industry-standard approximations with built-in tolerance bands. The contractual commitment defines the compliance threshold precisely; there is no universal industry-standard deviation allowance.